SetuCredit: Borrower Readiness Index (BRI)

Alternative Underwriting Architecture, Authoritative DPI Data Sources, Comparative Bureau (CIBIL) Analysis, and Financial Inclusion for India's 300 Million Unbanked Micro-Borrowers

Executive Summary: Traditional credit bureaus (CIBIL, Experian, CRIF, Equifax) rely exclusively on past debt repayment track records, penalizing informal rural workers, smallholder farmers, and micro-entrepreneurs with a default **'No History' (-1 / 0)** rating. SetuCredit's **Borrower Readiness Index (BRI, 0–100)** replaces manual, paper-intensive underwriting with an automated, explainable index computed in real time from India's Digital Public Infrastructure (DPI) — combining Account Aggregator (AA) cashflows, state land registries, utility payments (DISCOM), Bharat BillPay (BBPS), and the RBI Unified Lending Interface (ULI). Operating strictly as an in-memory pass-through layer, SetuCredit acts as the bridge that turns previously invisible borrowers into prime, bankable citizens.

Part 1: The Credit Exclusion Paradox & Bureau Failure

In India, over 300 million adults operate outside the formal banking perimeter. While they lack credit cards and formal term loans, they possess continuous, verifiable economic activity: paying electricity bills every month, cultivating title-cleared agricultural land, executing micro-transfers via UPI, and maintaining cellular recharges. Traditional credit models fail to capture this reality due to structural architectural assumptions:

- **The Catch-22 of Formal Credit:** To obtain a formal loan, a borrower must have a credit score; but to build a credit score, they must have already taken and repaid a formal loan. First-time micro-borrowers remain permanently locked out.
- **Lagged Batch Reporting:** Bureau scores update in 30-to-45 day batch reporting cycles from member institutions, making real-time instant micro-lending impossible.
- **Asymmetrical Information Cost:** Verifying an unbanked borrower through physical field visits, patwari land records, and manual inspection costs lenders ■1,500–■3,500 per file — uneconomical for ■25,000 to ■100,000 micro-ticket sizes.
- **High Vulnerability to Informal Lenders:** Denied formal credit, micro-borrowers resort to unregistered moneylenders charging 36% to 120% APR, triggering generational cycles of rural debt distress.

Part 2: How BRI (Borrower Readiness Index) is Calculated

The Borrower Readiness Index (BRI) is an explainable index bounded within **[0, 100]**, executed in a dedicated scoring microservice (`scoring/model/bri.py`). The model synthesizes 5 DPI data pillars, each assigned a deterministic point weight representing maximum attainable points:

Pillar & Source ID	Weight	Max Points	Regulatory / Tech Rail	Core Extracted Behavioral Features
Account Aggregator (aa)	25%	25 pts	Sahamati / RBI NBFC-AA	Average balance (norm ■50k), inflow consistency, account vintage (up to 60 mos), existing EMI burden ratio
Land Registry (land)	20%	20 pts	State Portals (Dharani, Bhulekh)	Agricultural parcel area (norm 5 acres), verified unencumbered title, years parcel held (norm 20 yrs)
Utility / Power (discom)	20%	20 pts	State Electricity Boards	On-time payment ratio, consecutive billing history (norm 24 mos), monthly consumption tier (norm ■3,000)
Telecom & Bills (bbps)	20%	20 pts	NPCI Bharat BillPay	Prepaid recharge consistency, SIM active vintage (norm 24 mos), monthly recharge tier (norm ■500)
Credit & History (uli)	15%	15 pts	RBI Unified Lending Interface	Existing active loan burden (cap 4), regular repayment history (norm 24 mos), past 12-month delinquencies

Sub-Component Normalization & Mathematical Formulations

Each raw payload maps to an in-memory feature vector (`scoring/features/extract.py`). Subscores $C_s \in [0, 1]$ are computed via monotonic clamping $\text{clamp}(x) = \max(0, \min(1, x))$:

Land: `clamp(0.50*(acres/5) + 0.30*clear + 0.20*(years/20))`	**DISCOM:** `clamp(0.60*on_time + 0.20*(mos/24) + 0.20*(bill/3000))`
AA (Bank): `clamp(0.40*(bal/50k) + 0.40*inflow + 0.20*(mos/60) - 0.30*emi)`	**BBPS:** `clamp(0.50*consistency + 0.30*(mos/24) + 0.20*(recharge/500))`
ULI: `clamp(0.50*(1 - loans/4) + 0.30*(repay/24) + 0.20*(1 - delinq/3))`	**Aggregated Score:** `BRI = round( sum( 100 * Weight_s * C_s ) )`

Zero-Inflation Policy: Every source is worth $100 \times W_s$ fixed points. Absent or withheld sources contribute 0 points rather than renormalizing remaining weights, eliminating consent-withholding gaming. Lender policy requires minimum mandatory consent for `aa` and `uli` to run an appraisal.

Readiness Tiers & Indicative Micro-Loan Sizing

Score Tier	Readiness Band	Indicative Loan Band	Underwriting Policy & Action
80 – 100	**Strong**	₹1,00,000 – ₹5,00,000	Instant straight-through processing (STP) disbursal via partner NBFC over UPI/IMPS rails.
60 – 79	**Good**	₹50,000 – ₹2,00,000	Standard algorithmic approval; recommended for seasonal agricultural working capital.
40 – 59	**Fair**	₹25,000 – ₹75,000	Micro-credit starter loan. Fast repayment lifts borrower to 'Good' within 6 months.
< 40	**Building**	*None (Below floor)*	Declined for immediate lending. UI generates automated advisory on lifting electricity regularity and balances.

Part 3: Deep Comparative Analysis — BRI vs. CIBIL Bureau

Dimension	Traditional Bureau (CIBIL)	SetuCredit BRI
Primary Purpose	Assess default probability on formal credit lines based on past credit debt.	Evaluate operational viability & income readiness for thin-file / unbanked borrowers.
Score Range	300 to 900 (Industry prime threshold typically >= 750).	0 to 100 (Explainable multi-source index with calibrated micro-loan bands).
Handling of New-To-Credit (NTC)	Returns **-1 or 0 (NH: No History)**. Result: Automatic rejection or predatory rates.	Evaluates non-financial operational records. Produces a full score even with zero prior loans.
Data Sources	Exclusively financial institutions: credit cards, term loans, mortgages, overdrafts.	DPI rails: Land registries (Dharani/Bhulekh), DISCOM utilities, BBPS telecom, AA bank flows.
Data Ingestion Model	Batch reporting: Lenders upload files monthly (30–45 day latency).	Real-time parallel API fan-out: Synchronous DPI pull completed in under 5 seconds.
Data Privacy & Storage	Perpetual data warehouse storing decades of loan, balance, and default history.	**Strict pass-through**: Non-financial data processed in RAM and immediately dropped. Never persisted.
Consent & Regulation	CIC Act; blanket consent signed at loan application.	DPDP Act compliant: Explicit, granular, time-bound (900s TTL), instantly revocable.
Explainability	Proprietary black-box algorithm; adverse action notices provide generic codes.	100% transparent point attribution: Borrower sees exact points earned per DPI pillar.
Cost per Appraisal	Direct bureau pull fees plus massive field verification / document collection overhead.	Zero field paperwork. Automated DPI fetch costs a fraction of physical verification.
Inclusivity Focus	Biased towards formal, salaried urban workers with documented income statements.	Designed natively for rural smallholders, SHGs, gig workers, and micro-entrepreneurs.

Part 4: Authoritative Data Extraction Sources & Technical Rails

SetuCredit extracts features exclusively from authoritative government, regulatory, and public digital rails. No unverified, self-reported, or third-party web-scraped data enters the pipeline. Below is the complete source-level mapping:

Pillar & Source ID	Governing Authority & Rail	Primary Query Keys	Exact In-Memory Extracted Schema	Underwriting Significance

State Land Registries
`source_id: land`
	State Revenue Depts
	• Dharani (Telangana)
	• Bhulekh (UP, Odisha, Bihar)
	• Bhoomi (Karnataka)
	• MahaBhulekh (MH)
		• Cadastral Survey No.
		• District / Mandal / Tehsil
		• Village Code / Khatauni
			`area_acres (float)`
			`ownership (clear|disputed)`
			`years_held (int 1-30)`
			`registry (str)`
				Validates immovable asset-backing and rural livelihood stability. Clear tenure eliminates mortgage registration costs for micro-loans.

Electricity Utilities
`source_id: discom`
	Ministry of Power / DISCOMs
	• State Utilities (e.g. TSSPDCL, UPPCL, BESCOM, MSEDCL)
	• Urja Mitra / NPP API
		• Consumer CA Number
		• Service Connection ID
		• Aadhaar-linked Mobile
			`months_paid (int 6-24)`
			`on_time_ratio (float 0-1)`
			`avg_monthly_bill_inr (int)`
			`utility (str)`
				24-month recurring utility payments serve as an ironclad proxy for payment discipline and household cashflow stability.

Account Aggregator
`source_id: aa`
	Reserve Bank of India
	• Sahamati Ecosystem
	• Licensed NBFC-AAs (Setu, Anumati, Finvu, OneMoney)
	• FIPs: Commercial/RRBs
		• AA Handle (`mobile@aa`)
		• Time-bound DPDP Consent Token
		• Linked Bank FIP Account
			`avg_balance_inr (int)`
			`inflow_consistency (float)`
			`months_active (int 6-60)`
			`existing_emi_ratio (float)`
				Evaluates true transaction velocity, seasonal crop/cash deposits, and existing debt outflows without requiring ITRs or salary slips.

Bharat BillPay
`source_id: bbps`
	NPCI Bharat BillPay
	• Central BBPS Switch
	• Telecom Operators (Jio, Airtel, Vi, BSNL)
	• Gas / DTH / Utility Billers
		• Subscriber Mobile MSISDN
		• Biller Consumer Number
		• Biller Operating Unit (BOU)
			`months_recharged (int 3-24)`
			`recharge_consistency (float)`
			`avg_monthly_recharge_inr`
			`biller (str)`
				The essential 'thin-file footprint'. Continuous prepaid recharges on a single SIM demonstrate location stability and disposable income.

Unified Lending Interface
`source_id: uli`
	Reserve Bank Innovation Hub
	• RBI ULI National Gateway
	• Non-prod: `extgw.nonprod.rbihub.io`
	• Production ULI Rail
		• Borrower Session Ref
		• DPDP Consent Certificate
		• Mutual TLS + HMAC-SHA256
			`existing_loans (int 0-4)`
			`repayment_regular_months`
			`delinquencies_12m (int)`
			`record_count (int)`
				Captures historical priority-sector credit, Kisan Credit Card (KCC) history, and SHG bank linkage loans missed by traditional bureaus.

Identity & Verification
`gate: identity`
	UIDAI & MeitY
	• Licensed ASA/AUA Gateway
	• DigiLocker eKYC API
		• Aadhaar registered mobile
		• 6-digit one-time passcode
			`txn_id (str)`
			`mobile_verified (bool)`
			`masked_uid (str, demo)`
				Proves session ownership without collecting or storing the raw 12-digit Aadhaar number (UIDAI regulatory compliance).

Part 5: Bridging the Credit Gap Using Operational Surrogates

A common misconception is that unbanked borrowers lack data. They do not lack data — they lack **structured aggregation**. India's DPI stack provides authoritative, tamper-proof operational records that act as direct mathematical surrogates for creditworthiness:

- **Electricity Bills as Debt-Service Surrogate:** A consumer who pays their DISCOM electricity bill on time every month for 24 months has established an ironclad habit of debt servicing. Power bills are recurring obligations with severe consequences for non-payment (disconnection), making payment discipline directly predictive of loan repayment.
- **Prepaid Mobile Recharges as Disposable Income Proxy:** India's rural informal economy runs on prepaid cellular connectivity. Consistent monthly recharges on a single mobile number over 12–24 months prove personal stability, location continuity, and steady disposable micro-cashflow.
- **Account Aggregator as Alternative Income Statement:** Micro-borrowers rarely possess formal payslips or audited balance sheets. Account Aggregator (AA) rails analyze digital transaction velocity, daily UPI inflows, Jan Dhan deposits, and seasonal crop sale receipts to calculate actual debt-servicing capacity.
- **State Land Registries as Collateral Anchor:** Land ownership records verified via state portals (e.g., Dharani, Bhulekh) provide definitive proof of rural asset ownership. Verifying parcel size, unencumbered status, and historical tenure gives lenders confidence without requiring mortgage deeds.

The Day-0 to Prime Credit Ladder:
BRI does not attempt to displace credit bureaus permanently. Instead, it serves as the **Day-0 onboarding catalyst**. When an NTC borrower is approved for an initial ■50,000 working capital loan via SetuCredit, the partner NBFC reports subsequent repayments to CIBIL and CRIF. Within 12 months, the borrower establishes a formal bureau score of 750+, graduating into mainstream banking.

Expansion Rails: Filling the Gap with Additional Operational Feeds

SetuCredit's modular adapter architecture (`backend/app/dpi/base.py`) enables rapid integration of emerging public digital registers without modifying scoring consumer interfaces:

- **Dairy & Agri-Cooperatives (NDDB/PACS):** Daily milk pour records and cooperative payment logs act as verified weekly cashflow proofs.
- **GSTN & Trade Invoicing:** Micro-merchant invoices and e-way bills underwrite rural kirana stores and trade supply chains.
- **PM-KISAN & Direct Benefit Transfers (DBT):** Verifiable government agricultural support payouts prove subsidy stability.
- **DigiLocker Vehicle Records:** Commercial tractor, auto-rickshaw, and light commercial vehicle RC proofs anchor mobility loans.

Part 6: Architecture, Compliance & Pass-Through Trust Boundary

To protect rural borrowers from predatory profiling and comply strictly with India's Digital Personal Data Protection (DPDP) Act, SetuCredit enforces non-negotiable architectural constraints:

- **Strict Pass-Through by Design:** Non-financial borrower payloads (land acreage, electricity bills, bank statements) live strictly in ephemeral RAM for the duration of the scoring request. They are NEVER written to disk, PostgreSQL, or logs.
- **Audit Ledger Isolation:** PostgreSQL stores only tamper-proof metadata: event timestamps, record counts, consent references, and SHA-256 payload digests for regulatory verification.
- **Stateful Consent Gate:** Every DPI adapter call is guarded in code by `app.consent.service`. Calls without active, unexpired consent return immediate HTTP 403 Forbidden.
- **HMAC Signed Handoff:** Pre-underwritten appraisal packages are cryptographically signed with HMAC-SHA256 before webhook transmission to partner NBFCs, guaranteeing message integrity over public networks.

SetuCredit Whitepaper Series • Ref: HLD-SC-2026-V1 • Engineered with FastAPI, Redis, PostgreSQL, React, and Python ReportLab. Repository: https://github.com/nchary05-oss/SetuCredit